

Maximum of Second Order Concomitants of Selected Order Statistics: A Study in the Morgenstern Bivariate Exponential Framework

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Abstract

This paper analyses the finite-sample and asymptotic distribution theory for the maximum of second order concomitants related to selected order statistics from the Morgenstern-type bivariate exponential distribution. Furthermore the paper provides illustrative tables that assess the effectiveness of the selection procedure based on this statistic.

Keywords: Order statistics, Second order concomitants, Morgenstern family, Bivariate exponential, Selection procedure.